

DTF via Robust Scaling: FCI Example

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2026-08-06

Initialization

Data Prep

Merge item data (`item_file`) with demographics (`demo_file`), subset to Fall 2013 pretest with gender.

```
# Import demographic data
dat_demo <- readxl::read_excel(demo_file) |> data.frame()
dat_demo <- dat_demo[, c("PID", "STUDENT_GENDER_IPEDS")]

# Import FCI data, remove unnecessary columns, change PID to numeric & remove NA
dat_fci <- readr::read_csv(item_file) |> data.frame()
dat_fci <- dat_fci[, -c(1:2)]
dat_fci$PID <- as.numeric(dat_fci$PID)
dat_fci <- dat_fci[!is.na(dat_fci$PID), ]

# Merge demographic data with FCI
dat_merge <- left_join(dat_fci, dat_demo, by="PID")

# Subset to Fall 2013 pre-test
dat_merge <- dat_merge[dat_merge$Pre.or.Post == "PRE", ]
dat_merge <- dat_merge[!is.na(dat_merge$STUDENT_GENDER_IPEDS), ]
dat_merge <- dat_merge[!duplicated(dat_merge$PID), ]
dat_year <- dat_merge[dat_merge$Term == "Fall 2013", ]
table(dat_year$STUDENT_GENDER_IPEDS)
```

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Helper function for running LRT

This function runs LRT using two-stage purification and refinement with BH correction. Each item-level LRT tests slopes and intercepts jointly.

```
lrt_dif <- function(dat0, dat1, alpha = .05){
  dat.mg <- rbind(dat0, dat1)
  groups <- c(rep("g1", times = nrow(dat0)),
              rep("g2", times = nrow(dat1)))
  dif.par <- c("a1", "d")
  purify.const <- c("free_mean", "free_var", "slopes", "intercepts")
  fit.purify <- multipleGroup(
    dat.mg,
    model = 1,
    group = groups,
    invariance = purify.const)

  dif.purify <- DIF(fit.purify,
                    which.par = dif.par,
                    scheme = "drop")
  dif.purify$p[is.nan(dif.purify$p)] <- 1
  anchor <- rownames(dif.purify)[dif.purify$p > alpha]

  if(length(anchor) < ncol(dat.mg)){
    not.anchor <- colnames(dat.mg)[!colnames(dat.mg)%in%anchor]

    # New base model
    anchor.const <- c("free_mean", "free_var", anchor)
    logL.baseline <- multipleGroup(
      dat.mg,
      model = 1,
      group = groups,
      invariance = anchor.const)@Fit$logLik

    # Fit models that release constraints on each anchor item
    logL.anchor <-
      lapply(1:length(anchor),
            function(x){
              constraints <- c("free_mean", "free_var", anchor[-x])
              multipleGroup(
                dat.mg,
                model = 1,
```

```

        group = groups,
        invariance = constraints)@Fit$logLik
    }
)

# Fit models that add constraints to each non-anchor item
logL.not.anchor <-
  lapply(1:length(not.anchor),
    function(x){
      constraints <- c("free_mean", "free_var", anchor, not.anchor[x])
      multipleGroup(
        dat.mg,
        model = 1,
        group = groups,
        invariance = constraints)@Fit$logLik
    }
  )

LR <- -2 * c((logL.baseline - unlist(logL.anchor)),
             (unlist(logL.not.anchor) - logL.baseline))

LR[LR < 0] <- 0
p.vals <- 1 - pchisq(LR, length(dif.par))
p.adj <- p.adjust(p.vals, method = "BH")
dif.table <- data.frame(
  items = c(anchor, not.anchor),
  chi.square = LR,
  p.vals = p.vals,
  p.adj = p.adj)
list(dif.items = c(anchor, not.anchor)[
  p.adj < alpha],
  dif.table = dif.table[order(dif.table$items), ],
  dif.par = dif.par)
} else {
  list(dif.items = NULL, dif.table = NULL, dif.par = dif.par)
}
}

```

Testing for DTF

Testing for DTF, using robust scaling and the likelihood ratio test or DIF.

```

# DTF Using Robust scaling
items <- grep("Score", names(dat_year))
dat_female <- dat_year[dat_year$STUDENT_GENDER_IPEDS == "F", items]
fit_female <- mirt(dat_female,
  model = 1,
  itemtype = "2PL",
  SE = T,
  verbose = F)

dat_male <- dat_year[dat_year$STUDENT_GENDER_IPEDS == "M", items]
fit_male <- mirt(dat_male,
  model = 1,
  itemtype = "2PL",
  SE = T,
  verbose = F)

mle <- get_model_parms(list(fit_female, fit_male))
rdif.out <- rdif(mle, fun = "d_fun3", alpha = .05, alpha.adjust = "BH")
rdif.delta <- rdif.out$delta.test

# DTF Using LRT
dat0 <- dat_female
dat1 <- dat_male

lrt.dif <- lrt_dif(dat_female, dat_male)
lrt.dif.items <- which(colnames(dat_female)%in%lrt.dif$dif.items)
lrt.delta <- robustDIF:::delta_test_from_dif(
  lrt.dif.items,
  mle,
  fun = "d_fun3")

# Summary of DTF tests (Table 3 from MS)
names(rdif.delta) <- names(lrt.delta)
dtf.tab <- bind_rows(
  RDIF = rdif.delta,
  LRT = lrt.delta,
  .id = "method") |>
  mutate(across(where(is.numeric), ~ round(.x, 3)))
#xtable::xtable(dtf.tab, digits = 3)
dtf.tab

```

```

method naive.est naive.se dif.est dif.se delta delta.se z.test p.val

```

1	RDIF	0.997	0.087	0.876	0.092	0.120	0.042	2.838	0.005
2	LRT	0.997	0.087	0.938	0.089	0.058	0.023	2.587	0.010

Testing for DIF

Follow-up with item-level tests to identify source of DTF.

```

item.descriptors <- c("Dropping unequal metal balls","Rolling balls off table","Why objects a

# Summary of item level tests (Table 4 from MS)
rdif.dif <- rdif.out$dif.test
if (nrow(rdif.dif) != ncol(dat_female)) {
  stop("RDIF item-level test output does not match the number of FCI items.")
}

rdif.dif.items <- colnames(dat_female)[rdif.dif$p.val < .05]

dif.tab <- cbind(item.descriptors,
  round(cbind(rdif.dif, lrt.dif$dif.table[, -1]), 3))

#xtable::xtable(dif.tab, digits = 3)
dif.tab

```

	item.descriptors	delta	se	z.test	p.val
item1_d1	Dropping unequal metal balls	0.372	0.174	2.141	0.032
item2_d1	Rolling balls off table	-0.043	0.174	-0.248	0.804
item3_d1	Why objects accelerate downward	0.191	0.184	1.037	0.300
item4_d1	Truck<U+2013>car collision forces	-0.111	0.182	-0.608	0.543
item5_d1	Forces inside curved channel	-0.412	0.207	-1.991	0.046
item6_d1	Ball exiting curved track	0.833	0.218	3.825	0.000
item7_d1	String breaks during swing	-0.021	0.156	-0.136	0.892
item8_d1	Kicked moving hockey puck	0.120	0.159	0.755	0.450
item9_d1	Speed after sideways kick	-0.037	0.284	-0.129	0.898
item10_d1	Speed after impulsive kick	0.072	0.106	0.679	0.497
item11_d1	Forces on sliding puck	-0.392	0.176	-2.224	0.026
item12_d1	Cannonball off a cliff	0.638	0.194	3.294	0.001
item13_d1	Thrown ball in midair	-0.165	0.139	-1.183	0.237
item14_d1	Object dropped from airplane	0.810	0.184	4.394	0.000
item15_d1	Pushing truck while accelerating	0.061	0.226	0.267	0.789
item16_d1	Pushing truck at constant speed	0.041	0.152	0.267	0.789
item17_d1	Elevator moving at constant speed	-0.161	0.241	-0.669	0.503

item18_d1	Forces on swinging child	-0.496	0.182	-2.730	0.006
item19_d1	Comparing block speeds	-0.073	0.180	-0.405	0.685
item20_d1	Comparing block accelerations	0.146	0.139	1.054	0.292
item21_d1	Spaceship with sideways thrust	0.670	0.271	2.473	0.013
item22_d1	Spaceship speed under thrust	0.312	0.183	1.699	0.089
item23_d1	Spaceship after engine cutoff	0.777	0.157	4.940	0.000
item24_d1	Post-thrust spaceship speed	0.374	0.131	2.852	0.004
item25_d1	Pushing box at constant speed	-0.133	0.200	-0.665	0.506
item26_d1	Doubling push on box	-0.124	0.285	-0.436	0.663
item27_d1	Stopping applied force	0.786	0.281	2.803	0.005
item28_d1	Students pushing off chairs	0.176	0.123	1.437	0.151
item29_d1	Forces on resting chair	-0.439	0.199	-2.202	0.028
item30_d1	Forces on flying tennis ball	-0.160	0.186	-0.862	0.389

chi.square p.vals p.adj

item1_d1	3.387	0.184	0.325
item2_d1	2.545	0.280	0.400
item3_d1	1.491	0.474	0.538
item4_d1	1.738	0.419	0.524
item5_d1	7.727	0.021	0.070
item6_d1	12.310	0.002	0.016
item7_d1	11.842	0.003	0.016
item8_d1	2.593	0.274	0.400
item9_d1	6.594	0.037	0.092
item10_d1	1.282	0.527	0.564
item11_d1	10.628	0.005	0.025
item12_d1	6.622	0.036	0.092
item13_d1	5.455	0.065	0.151
item14_d1	12.368	0.002	0.016
item15_d1	0.858	0.651	0.674
item16_d1	7.900	0.019	0.070
item17_d1	1.528	0.466	0.538
item18_d1	12.819	0.002	0.016
item19_d1	2.868	0.238	0.397
item20_d1	0.052	0.974	0.974
item21_d1	9.899	0.007	0.030
item22_d1	5.129	0.077	0.154
item23_d1	32.792	0.000	0.000
item24_d1	4.002	0.135	0.254
item25_d1	2.002	0.367	0.479
item26_d1	2.643	0.267	0.400
item27_d1	5.256	0.072	0.154
item28_d1	1.451	0.484	0.538
item29_d1	7.243	0.027	0.080

```
item30_d1      2.277  0.320 0.437
```

Comparison with Raju's DTF

Raju's DTF is computed here using `mirt::DTF()` applied to a two-group `multipleGroup()` model. Item parameters are placed on a common scale by constraining the RDIF-selected anchor items to be equal across groups while freeing the latent mean and variance.

```
raju.anchor <- colnames(dat_female)[!colnames(dat_female) %in% rdif.dif.items]
if (!length(raju.anchor)) {
  stop("The Raju DTF comparison requires at least one anchor item.")
}

dat_raju <- rbind(dat_male, dat_female)
group_raju <- factor(
  c(rep("Male", nrow(dat_male)),
    rep("Female", nrow(dat_female))),
  levels = c("Male", "Female"))

fit_raju <- multipleGroup(
  dat_raju,
  model = 1,
  itemtype = "2PL",
  group = group_raju,
  invariance = c("free_mean", "free_var", raju.anchor),
  SE = TRUE,
  verbose = FALSE)

raju.dtf <- mirt::DTF(fit_raju)

raju.dtf.tab <- data.frame(
  statistic = names(raju.dtf),
  value = as.numeric(raju.dtf))

raju.dtf.tab |>
  mutate(value = round(value, 3))
```

```
      statistic value
1 sDTF.score 0.400
2 sDTF(%).score 1.333
3 uDTF.score 0.409
```

4 uDTF(%) .score 1.362

```
mirt::DTF(fit_raju, draws = 500, plot = 'func')
```

